



Postoperative radiotherapy in primary resectable desmoid tumors of the neck: a case–control study

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Received: 13 March 2019 / Accepted: 20 May 2019 / Published online: 6 June 2019
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Abstract

Purpose Desmoid tumors (DTs) are rare soft tissue tumors, which frequently recur after surgery. The optimal treatment approach is still controversial due to lack of randomized data. The purpose of the study was to review a relative large series of DTs of the neck with emphasis on the value of postoperative radiotherapy (RT).

Methods A total of 68 patients with DTs of the neck region with or without postoperative RT between 1/2008 and 12/2017 were included. They were individually matched for factors including age, gender, tumor size and margin status with a cohort of patients who underwent operative without postoperative RT. The event-free survival (EFS) was compared in the two groups.

Results For the study, 34 patients and 34 matched control subjects were identified. The two groups did not differ in terms of age ($p=0.810$), gender ($p=0.328$), tumor size ($p=0.803$) and margin status ($p=0.799$). Patients who received surgery plus RT had a significantly higher 3-year EFS than those who received surgery alone (74.6% vs. 13.3%, $P<0.001$). Factors including postoperative RT and margin status were independent factors of EFS. The values of P were <0.001 and 0.003 and the hazard ratios (95% confidence interval) were 11.929 (4.732–30.072) and 0.299 (0.133–0.671), respectively. None of the patients treated with RT developed serious complications.

Conclusions According to our results, postoperative radiotherapy is an effective treatment in desmoid tumors of the neck. Multi-institutional and prospective studies are warranted to further validate our findings.

Keywords Soft tissue neoplasms · Fibromatosis, aggressive · Head and neck neoplasm · Resection margins · Benign neoplasms

Postoperative Strahlentherapie bei primär resezierbaren Desmoidtumoren im Halsbereich: eine Fallkontrollstudie

Zusammenfassung

Zielsetzung Desmoidtumore (DTs) sind seltene Weichgewebetumoren, die nach der Operation häufig wieder auftreten. Die optimale Behandlung von DTs bleibt wegen des Mangels an randomisierten Daten umstritten. Ziel der Studie war es, eine relativ große Serie der DTs im Halsbereich mit Schwerpunkt auf den Nutzen der postoperativen Strahlentherapie (RT) zu überprüfen.

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Methoden Insgesamt wurden Patienten mit DTs im Halsbereich mit oder ohne postoperative RT zwischen 1/2008 und 12/2017 eingeschlossen. Sie wurden individuell auf Faktoren wie Alter, Geschlecht, Tumorgröße und Resektionsrand (R-Status) mit einer Kohorte von Patienten abgestimmt, die eine operative ohne postoperative RT durchliefen. Das ereignisfreie Überleben (EFS) wurde in zwei Gruppen verglichen.

Ergebnisse Für die Studie wurden 34 Patienten und 34 aufeinander abgestimmte Kontrollpersonen identifiziert. Die beiden Gruppen unterschieden sich nicht in Bezug auf Alter ($p=0,810$), Geschlecht ($p=0,328$), Tumorgröße ($p=0,803$) und R-Status ($p=0,799$). Patienten, die eine Operation plus RT erhielten, hatten ein deutlich höheres dreijähriges EFS als diejenigen, die eine Operation allein erhielten (74,6% vs. 13,3%; $P<0,001$). Faktoren wie postoperative RT und R-Status waren unabhängige Faktoren von EFS. Die Werte von P waren $<0,001$ und $0,003$ und das Hazard-Verhältnis (95 %-Konfidenzintervall) betrug jeweils 11,929 (4,732–30,072) bzw. 0,299 (0,133–0,671). Keiner der Patienten, die mit RT behandelt wurden, entwickelte ernsthafte Komplikationen.

Schlussfolgerung Nach unseren Ergebnissen ist die postoperative Strahlentherapie bei DTs im Halsbereich wirksam. Multiinstitutionelle und prospektive Studien sind gerechtfertigt, um unsere Ergebnisse weiter zu validieren.

Schlüsselwörter Weichteilneoplasie · Aggressive Fibromatose · Kopf-Hals-Neoplasie · Resektionsrand · Gutartige Neoplasien

Introduction

Desmoid tumors (DTs), also known as aggressive fibromatosis (AF), were defined by the World Health Organization (WHO) as a “clonal fibroblastic proliferation that arises in the deep soft tissues” [1]. It is a distinct rare entity with an incidence rate of approximately five per million people annually [2]. DTs are summarized as benign tumors, but the term benign is deceptive because it fails to highlight the locally invasive pattern and high local recurrence rates after local resection [3, 4].

DTs can occur and develop in a wide variety of anatomic locations and are classified into three types: intra-abdominal, abdominal, and extra-abdominal types. Head and neck DTs are one of the extra-abdominal DTs and estimated to constitute only 7 to 15% of all DTs throughout the body [5, 6]. Among the head and neck DTs, the neck is the most common site of occurrence [7, 8]. Because of the low incidence, the experience of the neck DTs is limited to small series and case reports. Only a few head and neck series have reported more than 20 patients [7, 9–12].

According to the recommendations of a recent European Consensus, the ‘watchful waiting’ approach is the standard treatment for asymptomatic patients of DTs [13]. As a rare disease, the successful management of DTs is undertaken by a multidisciplinary team with sarcoma expertise according to the National Comprehensive Cancer Network (NCCN) Guidelines Version 2.2018 of the soft tissue sarcoma [14]. DTs are frequently recur after surgery. Other alternative treatment options like primary radiotherapy and systemic treatment can be used. There is no established treatment approach available for this disease. With respect to adjuvant radiation, the group of patients for whom this is suitable therapy remains a matter of debate. Therefore, the purpose of the study was to review a relative large series of

the neck DTs treated at our institution with an emphasis on the value of postoperative radiotherapy (RT).

Methods

Patient selection

The records of patients diagnosed with neck DTs and treated at Fudan University Shanghai Cancer Center from 1/2008 to 12/2017 were retrospectively reviewed. Desmoid histology was confirmed for all patients by experienced pathologists at our institution. The study group consisted of patients who underwent postoperative RT. They were individually matched for a number of factors including age, gender, tumor size and margin status with a cohort of patients who underwent operative without postoperative RT.

Therapeutic strategies

Surgical resection was done on all patients. Patients were retrospectively categorized into two groups for analysis according to the R-classification and margin status: group I (patients underwent complete tumor resection with negative margins [R0]), and group II (patients either underwent complete resection but with microscopic positive margin [R1], or underwent incomplete resection with macroscopic residual disease [R2]). Postoperative RT was delivered on the study group. RT techniques included 2-dimensional or 3-dimensional conformal or intensity modulated radiation therapy. RT doses varied from 44 to 68 Gy (median 60 Gy). Systemic treatment included chemotherapy and hormonal therapy, but were undertaken only in some patients who had local failures.

Patient assessments and follow-up

Given that overall survival (OS) is good following local treatment of DTs, we chose to focus on event-free survival (EFS). EFS considered local relapse or progressive disease (RECIST) after surgery with or without postoperative RT. Recurrence was defined as radiographic increase in tumor size after treatment. Radiotherapy-related toxicities were graded according to the Radiation Therapy Oncology Group (RTOG).

Statistical analysis

All data analyses were carried out using SPSS 22.0 (SPSS, Inc.). The chi-square (χ^2) test was used to test the differences between the study and control groups in terms of categorical variables. EFS curves were calculated using the Kaplan–Meier method from the date of diagnosis until the occurring event or last follow-up, and comparisons between curves were made by the log-rank test. Univariate analysis was conducted using the Kaplan–Meier method and multivariate analysis was conducted by Cox regression analysis. Parameters identified as significantly correlated with EFS in the univariate analysis were further tested by multivariate analysis. All statistical tests were two-sided, and *p*-values of 0.05 were considered statistically significant.

Results

The clinical characteristics of the case–control study

For the study, a total of 34 patients with DTs of the neck region who received surgery plus RT and 34 matched control subjects who received surgery alone were identified. The patients' characteristics are summarized in Table 1. The median follow-up time was 38.0 months (range 2–146 months) and 36.5 months (range 5–126 months) in the study and the control groups. The two groups did not differ in terms of

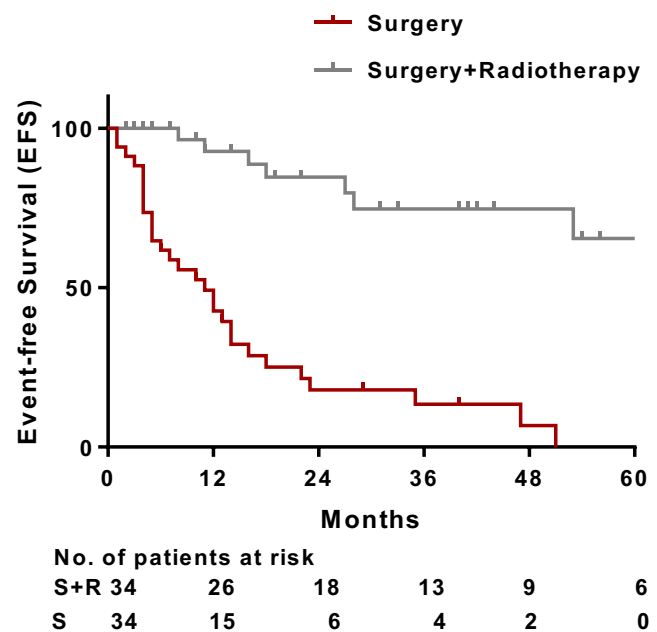


Fig. 1 Kaplan–Meier estimators of event-free survival (EFS) for the patients having surgery plus radiotherapy and in those receiving surgery alone

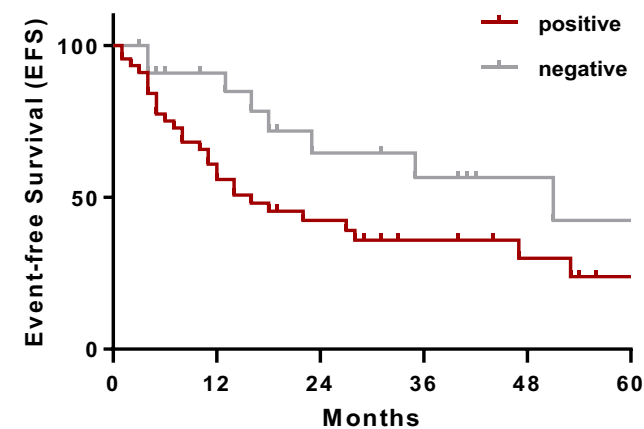
age ($p=0.810$), gender ($p=0.328$), tumor size ($p=0.803$) and margin status ($p=0.799$).

Event-free survival

The interval from surgery to first recurrence in the patients having surgery plus RT and in those receiving surgery alone was 31.4 ± 25.49 and 13.5 ± 12.98 months, respectively ($p=0.001$). Patients who received surgery plus RT had a significantly higher 3-year and 5-year event-free survival (EFS) rate than those who received surgery alone (74.6% [95% confidence interval/CI, 56.8% to 92.4%] vs. 13.3% [95%CI, 0.56% to 26.0%] and 68.3% [95%CI, 42.172% to 88.428%] vs. 0% [95% CI, 0% to 0%], respectively, $P<0.001$) (Fig. 1).

Table 1 Clinical characteristics of patients in the two treatment groups

Characteristic	Group	Surgery plus radiotherapy		Surgery		<i>P</i> value
		No.	%	No.	%	
Gender	Male	16	47.1	12	35.3	0.328
	Female	18	52.9	22	64.7	–
Age (years)	>40	18	52.9	17	50	0.810
	≤40	16	47.1	17	50	–
Tumor size (cm)	>5	12	35.3	13	38.2	0.803
	≤5	22	64.7	21	61.8	–
Surgical margin status	Positive	23	67.6	22	64.7	0.799
	Negative	11	32.4	12	35.3	–



No. of patients at risk						
positive	45	24	15	9	6	3
negative	23	16	10	8	4	4

Fig. 2 Kaplan–Meier estimators of event-free survival (EFS) for the patients with margin positive and negative

Univariate and multivariate analyses

Age, gender and tumor size had no significant impact on EFS. The 3-year EFS in the group of with negative margins ($n=23$) as compared to the patients with positive margins ($n=45$) were 42.2% vs. 35.9% ($p=0.054$, Fig. 2). Factors including postoperative RT and margin status were independent factors of EFS. The values of P were 0.000 and 0.003 and the hazard ratio (95%CI) were 11.929 (4.732–30.072) and 0.299 (0.133–0.671) in the factors of postoperative RT and margin status (Table 2).

Treatment compliance

Overall, complications of surgery including edema, neck pain and fibrosis were similar in the two groups and were well tolerated. The regimen of surgery plus RT was tol-

erated. There were no grade 4/5 complications. The most common radiation-related acute side effects were skin effects. Grade 1 acute dermatitis occurred in 20, grade 2 occurred in 6, and grade 3 occurred in 2 of the 34 patients.

Discussion

DTs are rare benign tumors, especially in the head and neck region. Apart from the research of Hoos et al. (21 cases) [7], Conley et al. (40 cases) [9], Wang et al. (20 cases) [10], Fasching et al. (48 cases) [11] and Gnepp et al. (25 cases) [12], most studies reported less than 10 cases ([8, 15–18]; Table 3). Among the head and neck DTs, there were less studies in one lesion of the neck. We analyzed 68 neck DTs in the past 10 years, a relative large series of neck DTs with an emphasis on the value of postoperative RT. We exclude the influence of age, gender, tumor size and margin status and find that postoperative RT is effective in the desmoid tumors of the neck.

In the recommendation of European Consensus, ‘watchful waiting’ approach is the standard treatment for asymptomatic patients of DTs [13]. Also, there is no established treatment approach available. According to the NCCN Guidelines Version 2.2018 of the soft tissue sarcoma, surgery is primary treatment in resectable DTs [19–23]. Several retrospective studies have reported that compared to surgery alone, postoperative RT significantly improves local control, suggesting that postoperative RT could be considered in patients at high risk of local recurrence [24–28]. However, in the chest wall of DTs, postoperative RT did not reduce local recurrence [23]. The patients for whom postoperative RT is suitable therapy remains a matter of debate. The category is 2A and there are no studies focusing on the neck DTs in the NCCN Guidelines Version 2.2018. In our report, patients with the neck DTs who received surgery plus RT had a significantly higher

Table 2 Impact of prognostic factors on 3-year event-free survival (EFS) by univariate and multivariate analyses

Variable		No.	3-year EFS (%)	Univariate analysis		Multivariate analysis	
				95%CI	P value	HR (95%CI)	P value
Gender	Male	28	33.7	9.0–58.4%	0.553	–	–
	Female	40	26.6	7.6–45.6%	–	–	–
Age (years)	>40	35	35.9	15.9–55.9%	0.687	–	–
	≤40	33	20.2	2.1–38.3%	–	–	–
Tumor size (cm)	>5	25	48.6	27.0–70.16%	0.639	–	–
	≤5	43	36.2	18.0–54.4%	–	–	–
Surgical margin status	Positive	45	35.9	20.4–51.4%	0.054	0.299 (0.133–0.671)	0.003
	Negative	23	42.4	11.8–73.0%	–	–	–
Group	Surgery+ Radiotherapy	34	74.6	56.8–92.4%	0.000	11.929 (4.732–30.072)	0.000
	Surgery	34	13.3	0.56–26%	–	–	–

Table 3 Series with DTs of the head and neck region in adults with more than 10 patients

Authors	Year	No. of patients	Treatment	Local failure (%)
Conley et al. [9]	1966	40	Surgery 40	47
Fasching et al. [11]	1988	48	Surgery 38 Surgery + RT 8 RT 2	46
Gnepp et al. [12]	1996	25	Surgery 24 Surgery + RT + Chemotherapy 1	21
Hoos et al. [7]	2000	21	Surgery 10 Surgery + RT 11	24
Wang et al. [10]	2006	24	Surgery 21 Surgery + RT 1 Wait-and-see 2	13

RT radiotherapy

3-year EFS rate than those who received surgery alone (74.6% [95%CI, 56.8% to 92.4%] vs. 13.3% [95%CI, 0.56% to 26.0%], respectively, $P < 0.001$). Postoperative RT appears to be significantly associated to local control and is recommended in neck DTs.

Tumor location, tumor size, age and margin status have been considered as factors associated with local recurrence following resection. Gronchi et al. [29] reported that patients with abdominal wall tumors had significantly higher DFS (disease-free survival) rates than those with extremity tumors, 88% and 62%, respectively ($p < 0.01$). Peng et al. [30] also reported similar findings. The 5-year DFS was 28.8% and 80.2% for patients with extremity tumors and abdominal wall tumors. However, the impact of tumor size, age and margin status on local control remains controversial. Some studies have reported tumor size [31], age [32], and margin status [30, 33–35] as independent prognostic factors of recurrence. Other studies have failed to demonstrate any clear association between tumor size [32, 36, 37], age [36, 37], margin status [25, 29, 36], and risk of recurrence. In our study of the neck DTs, there were no significant differences of the EFS in the factors age, gender, and tumor size. However, margin status was independent factors of EFS. The value of P was 0.003 and the hazard ratio (95%CI) was 0.299 (0.133–0.671) in the factor of margin status.

Treatment compliance, such as edema, neck pain and fibrosis, are the most prevalent as the results of surgery and RT with a potential contribution from the tumor itself. The most common radiation-related acute side effects were skin effects. In Bates' study [32], grade 3 or 4 toxicities occurred in 37% of patients because of the high recurrent disease (49.5%) and multiple operations (21.8%). Many treatment compliances were multifactorial in their study. The rate of toxicity in our report was lower in patients treated in the era of 2D RT [38]. The improvement of RT treatment techniques, such as intensity modulated radiation therapy, can

decrease the dose to adjacent normal tissue, thus reducing the risk of toxicity.

In conclusion, according to our results, radiation therapy is effective in the control of DTs of the neck region treated after surgery. Postoperative radiation therapy is a safe and effective treatment for desmoid tumors of the neck region. Multi-institutional and prospective studies are warranted to further validate our findings.

Conflict of interest X. Niu, R. Jiang and C. Hu declare that they have no competing interests.

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